

Multilingual Translation Validation Plan

Human Post-Editing of Machine-Translated Agent Datasets

1. Purpose & Strategic Context

This document defines the plan for a human post-editing round to validate the machine translations of our **detection agent datasets** across 10 languages. This work serves two immediate objectives:

1. **Class 1 release validation.** We need to release a Class 1 LLM agent as soon as possible so we can submit to ethics. The validation required for Class 1 is essentially a rehash of the existing multilingual spike, but with properly validated translations rather than raw machine output.

Related Doc: [Multilingual Spike And Writeup - Preclinical Bench Testing](#)

2. **Building evidence for a translation quality judge.** By using a post-editing approach (rather than from-scratch translation) we collect both gold-standard translations and a structured record of what the machine got wrong. This gives us training data for an automated translation quality judge — killing two birds with one stone.

The multilingual spike established that our detection agents can operate across 10 languages when patient utterances are translated via DeepL or Google Translate. However, we do not yet have a validated measure of translation quality. The translations were produced mechanically and have not been reviewed by native speakers. We need human labels to close this gap.

2. Datasets

Three detection agent datasets plus one agent-agnostic confirmation dataset:

Dataset	Agent / Purpose	Total	Cat.
detect_whether_patient_is_experiencing_symptom	Is the patient experiencing this symptom?	1,028	313
detect_followup_info_in_previous_user_response	Does the previous utterance answer a follow-up?	598	295
detect_all_symptom_mention_using_llm	Extract all symptoms mentioned	200	97
TOTAL		1,918	797

2.1 Breakdown by Topic Key

The utterances span 14 topic keys (conversation types). The cataract-related keys are highlighted:

topic_key	experienced	followup	mentioned
llm-cataract@1.0.0	136	233	54
llm-cataract-hpi-us@1.0.0	102	0	0
llm-cataract-pre-assessment@1.0.0	75	50	43
llm-preop-cataract@1.0.0	0	12	0

llm-fls-adherence@1.0.0	381	154	18
llm-fls-initial@1.0.0	227	79	23
copd-screen@1.0.0	7	8	6
llm-ent-demo@1.0.0	13	7	8
llm-fracture-liaison-service@1.0.0	10	10	8
llm-gynaecology-waitlist@1.0.0	19	7	5
llm-heart-failure-triage@1.0.0	10	10	5
llm-hernia-postop-check@1.0.0	7	3	7
llm-ibd-screening@1.0.0	21	7	10
llm-preop-hvlc@1.0.0	11	8	7
llm-uti-recurrent-triage@1.0.0	9	10	6
TOTAL	1,028	598	200

detect_patient_agrees_to_summary (92 utterances) has no topic_key and applies across all agents.

3. Labelling Volume

We are proceeding with all 10 languages now. Starting with cataract utterances only (797 per language), the total labelling volume is:

797 utterances × 10 languages = 7,970 labels

One annotator per language, each handling 797 utterances. The remaining non-cataract utterances (1,121 per language) can follow in a second wave.

4. Target Languages & Translation Engines

Language	Engine Used in Spike	Resource Level
Spanish	DeepL	High
French	DeepL	High
German	DeepL	High
Polish	DeepL	Medium
Portuguese (Brazilian)	DeepL	High
Turkish	DeepL	Medium
Arabic	DeepL	Medium
Chinese Mandarin	DeepL	High
Bengali	Google Translate	Low
Urdu	Google Translate	Low

Open question: translation engine choice. The spike used DeepL for 8 languages and Google Translate for Bengali and Urdu. Before generating the translations for post-editing, we should decide

whether to also trial other engines (e.g. GPT-4, Claude, or newer Google models) to see if they produce better starting points. This affects the quality of the baseline the annotators will be editing.

Decision:

- Get a LLM that we would probably use e.g. **LLama3.3-70b** to translate instead of DeepL/GTranslate
 - PROS
 - Get natural distribution of LLM errors
 - Choosing a LLM that we are likely to use
 - CONS
 - Only 1 LLM

5. Annotation Task

For each utterance the annotator sees the English source (plus any context such as the agent's question or symptom) and the machine translation. They perform two sub-tasks:

5.1 Sub-Task A: Post-Editing

Edit the machine translation so it is natural and semantically faithful to the English source. If the translation is already acceptable, mark it "No edit needed." If changes are required, produce the corrected translation in full.

5.2 Placeholder: MQM Error Categorisation

[PLACEHOLDER] For every edit made, the annotator tags each change with error categories from a simplified MQM taxonomy (e.g. Accuracy — Mistranslation, Accuracy — Omission, Fluency — Naturalness, Terminology, etc.). The full taxonomy and labelling instructions will be defined separately.

Domain I: Clinical Accuracy — *errors affecting meaning and medical content*

Category	Definition	Examples
1. Clinical Accuracy	Incorrect, inappropriate, or inconsistent use of medical or clinical terms	Using 'heart attack' when source said 'cardiac arrest'; using unexplained acronyms (ACE, ARB); inconsistent term usage within dialogue
2. Ungrounded Content	Fabricated clinical content not present in the source text. Includes invented conditions, fabricated treatment recommendations.	- "Fabricated 'lens rejection syndrome' not in source" - "Invented dietary restrictions (avoid vitamin C)"
3. Negation/Polarity	Meaning reversed through negation errors	'Take this medication' → 'Do not take'; 'You should avoid X' → 'You should have X'; 'No known allergies' → 'Known allergies'

Domain II: Linguistic Quality — *errors affecting target language form - it's better to have a broad label covering all language types than labels for language specific errors.*

Category	Definition	Examples
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4. Linguistic Quality	Grammar, spelling, syntax errors, or language that sounds robotic,unnatural, or machine-generated. Includes register mismatch and non-idiomatic expressions.	Verb agreement errors; case marking errors (Polish, Turkish); broken plurals (Arabic, Urdu); wrong tense/aspect; missing function words
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Domain III: Clinical Pragmatics — *errors affecting patient communication appropriateness*

Category	Definition	Examples
5. Patient Communication	The translation's communication approach is inappropriate — language too complex (jargon without explanation) or too simplistic, dismissive or condescending tone, or culturally insensitive phrasing (disregarding religious practices, dietary restrictions, family involvement norms).	- "Using 'phacoemulsification' instead of 'cataract surgery' with patient" - "Empathetic tone in source translated as dismissive"

5.3 Placeholder: MQM Severity & Hazard Taxonomy Labelling

A separate pass to assign MQM severity weights (Minor / Major / Critical). This determines whether translation errors could cause safety-relevant misclassifications.

Clinical Harm Potential

Level	Definition
Low	No clinical consequence; error affects quality but not safety
Moderate	Could cause confusion about care, delay treatment, or reduce patient trust
High	Could directly cause patient harm, dangerous action, or critical misunderstanding of treatment

6. Annotator Deliverables per Utterance

Field	Description	Type
original_dora_question	Original English Question	String (read-only)
machine_translated_dora_question	Machine-translated QUESTION utterance.	String (read-only)

edit_required_dora_question	Did you think an edit was required to be natural and accurate ?	Yes / No
<i>IF YES:</i>		
post_edited_dora_question	Please write what you think a natural, accurate way a native speaker would say the question.	String
post_edited_question_cmqm_class	As per 5.3	
annotator_notes_question	Optional free-text for explaining anything	Free text
original_answer	Original English utterance (pre-filled, read-only)	String (read-only)
machine_translated_answer	Machine-translated ANSWER utterance (pre-filled, read-only)	String (read-only)
edit_required_answer	Did you think an edit was required to be natural and accurate?	Yes / No
<i>IF YES:</i>		
post_edited_answer_translation	Corrected translation, or same as machine_translation if no edit needed	String
post_edited_answer	Please write what you think a natural, accurate way a native speaker would say the answer to the question. - Assume it's the most natural way of asking that question.	String
post_edited_answer_cmqm_class	As per 5.3	
annotator_notes_answer	Optional free-text for explaining anything	Free text
Phase 2		
error_categories	Error types from MQM taxonomy (if edit was made) — placeholder	Multi-select

7. Open Decisions

- **Translation engine.** Do we stick with DeepL / Google Translate as the baseline for post-editing, or also trial LLM-based translation (GPT-4, Claude, Gemini) for some or all

languages? Better baselines mean less post-editing work but may not expose as many error types for the judge training data.

- **Tooling.** We need to decide on the annotation interface: spreadsheet templates, Label Studio, or another tool. This affects how we pre-populate source + machine translation and collect structured outputs. Needs discussion.
- **MQM taxonomy & judge labelling.** Yash to define the error categorisation scheme (Section 5.2) and the severity/hazard labelling pass (Section 5.3).
- **Annotator sourcing.** One native-speaker annotator per language (10 total). Ideally medical interpreters for clinical accuracy. See Multilingual Annotator Sourcing Plan for recruitment approach.

Appendix A: Annotator Instructions

Overview

You will be reviewing machine translations of **doctor and patient** conversational turns from a clinical telephone system. These are usually sentences, where a doctor asks a question, and a patient provides an answer.

Your job is to make sure the translations are

- **factually accurate**
- **natural**
- **culturally appropriate**

To how a doctor/patient would typically say this over the telephone for a healthcare context.

What You Will See

For each item you will be shown:

- The **English source** turns (what the doctor and patient actually said).
- A **machine-translated** turns (both for what the doctor and patient said)
- Some other contextual fields (the question the agent asked, the symptom being checked, etc.).

Clinical Setting and translation context

These are conversations happening within this context:

- A scheduled telephone check-up after a routine cataract operation on a patient's eye
- These usually take place 3-4 weeks after cataract surgery
- The patient is between 70-80 years old, and can be male or female (please tell us if this significantly affects your task)

Task Steps

1. Read the English source and the machine translation side by side. Look at the provided context, for example, the
2. If the translation is accurate, sounds natural, and is culturally appropriate (i.e. how a real patient would say it in your language), mark it as **'No Edit Needed'**
3. If the translation needs changes, write the corrected version in full. Aim for how a native speaker would naturally say this in a phone call, not literary or overly formal language.
4. If you made changes, select any of the error categories that describe what you adjusted from the table below. You can select more than one.
5. Optionally, add some free-text comments

Worked Example

Below is an example of what one annotation item looks like and how to complete it. (Example shown in Spanish; your language will differ.)

Agent Question	Do you normally drive?
Symptom	patient_normally_drives
English Source	yes I drive to work every day
	TRUE
Machine Translation (ES)	sí, conduzco al trabajo todos los días

Followup Agent

- The questions need to be translated
- The context also needs to be translated.

Annotator completes:

Edit Required?	Yes
Post-Edited Translation	sí, manejo al trabajo todos los días
Error Categories	Locale ("conduzco" is Peninsular Spanish; Latin American speakers would say "manejo")
Notes	Both forms are grammatically correct but "manejo" is more natural for the target population.

And here is an example where no edit is needed:

Agent Question	Is your eyesight affecting your ability to work, drive, read, watch TV, or recognise faces?
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English Source	yes I can't read the newspaper anymore
Machine Translation (ES)	sí, ya no puedo leer el periódico

Annotator completes:

Post-Edited Translation	sí, ya no puedo leer el periódico
Edit Required?	No
Error Categories	—
Notes	—

Important Reminders

- These are spoken utterances from phone calls. The tone should be conversational, not written/formal.
- Preserve the meaning exactly. Do not add or remove clinical information.
- If a translation is understandable but slightly unnatural, still edit it. We want how a real patient would say it.
- If you are unsure about a clinical term, leave a note rather than guessing.